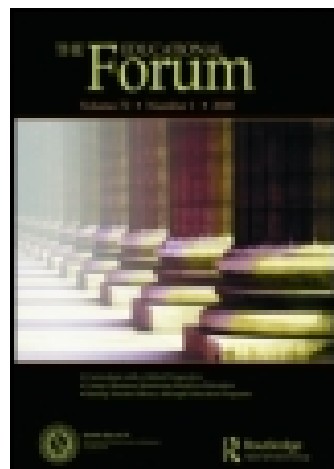


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Gifted Education: Needed Theory

Within the educational community there has long been a persistent core of individuals deeply concerned with the enhancement of the school experience for the most able of learners, the gifted.¹ The labors of these individuals have culminated in various programmatic changes for gifted learners and have stimulated many others within school and community to participate enthusiastically in the gifted movement. However, such optimistic statements must be tempered with a growing sense of concern.

Upon closer scrutiny it becomes apparent that much of what is being done under the banner of "gifted" is done without the knowledge of a theoretical framework to which it must relate. There appears to be a manipulation of surface elements within the gifted movement, without regard for the deeper structures or processes from which individual differences in abilities arise. As evidence of this, one can refer to the body of literature published within the category of the "gifted." This literature frequently purports to address some practical concerns in the education of more able learners, often without a sense of theory upon which various statements, suggestions, and proclamations can be rationalized. The results are writings that seem far too haphazard and random and come too frequently from the "it-worked-for-me-it-can-work-for-you" school of reasoning. Further, it must be questioned how practical such undertakings can be, without a clear link to a theory-base. In gifted education, where money and time are being invested in designing programs, developing better

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definitions, and devising new tests, far too little attention is being directed toward a greater understanding of the phenomenon to which these tests, programs, and definitions are supposedly related—the nature of individual differences in intelligence.

Intelligence, as broadly defined, represents “education’s most important product” and “its most important raw material.”² A greater understanding and more intense consideration of this construct should be at the heart of the gifted movement. Yet, in the rush to make education for the gifted more outwardly attractive, educators may be neglecting the theoretical basis on which that education is to be assembled, thus permitting the eventual collapse of all they have labored to develop.

Many of the developments in intelligence that have affected gifted education have come from researchers embracing clearly delineated theoretical paradigms. What is being addressed here is the lack of theoretical direction among those who have sought to apply the “tools” of these researchers. For example, while the instruments and empirical techniques of the psychometricians have been widely employed in gifted education, the psychometric theory-base has not permeated the ranks.³ Perhaps an alternative paradigm, with greater utility for practitioners and with more instructional adaptability, can significantly influence gifted education. It is the objective of this article to explore the domains of cognitive psychology and information processing as theoretical frameworks with the utility and adaptability needed to revitalize the gifted movement. The potential influence of an information-processing approach on gifted education will be examined with respect to two interrelated program issues: (1) identification, and (2) individualized instruction.

Identification

If you were to ask educators how they would identify a gifted individual, the characteristic that frequently tops the list is that of a high IQ or a high level of intelligence.⁴ But what is intelligence? For many practitioners, intelligence has been whatever the intelligence tests measured or whatever it was operationally defined to be. As Resnick put it,

For many years the term intelligence has been used by psychologists in a particular and very pragmatic sense to refer to the level of performance on tests designated intelligence tests.⁵

The psychometricians arrived at definitions of intelligence. Practitioners, in turn, followed the prescribed procedures, administered selected instruments, and identified their gifted (i.e., more intelligent) students. One consequence of this approach for gifted educators has been their over-reliance on scores on particular measures or tasks.

A second consequence has been the arbitrary demarcation of the boundaries of giftedness. Labels of "gifted" have or have not been attached to certain students solely as the result of insignificant variations in test performance. For example, while an IQ score of 130 may lead to an identification of giftedness, a score of 129 may warrant no such label. For both practitioners and psychometricians, it served little purpose to look beyond the individual performance on a particular instrument to question what internal operations may have produced such a performance. What was occurring in the individual that may have culminated in that specific score or performance was not of concern, and inadequate attention was given to the cultural or experiential influences that may have led the person to select one response over another. Regardless of the inclinations of some psychometricians, the measures they have produced are quantitatively analyzed and weighted, while qualitative information is relatively ignored. It has ultimately been the score that mattered not the process by which a score was achieved.

How would this behavior pattern differ if identification procedures for the gifted were based on cognitive theory? Should the identification of a gifted learner occur within the framework of cognitive psychology, we might expect variations from more standard procedures with regard to process, information, and response analyses.

Process Analysis

Building on the work of Carroll⁶ and Sternberg⁷, interest is developing in what various factors or component processes underlie specified information-processing tasks. In identification of gifted learners, this concern has resulted in the analysis of the processes required in the successful completion of selected identification measures or item types. For example, accurate response to linguistic or non-linguistic analogies (e.g., boy:girl:: man: _____, or _____ : _____ :: _____ : _____) make several process-demands on learners. These individuals must not only *encode* the separate meanings of each term, but also must *infer* the relationship between the paired terms (boy:girl), then generate an appropriate rule for that pairing (e.g., male/female; young in years). In turn, this inferred rule must be *mapped* onto the second, incomplete pairing, and a suitable response *applied* (man:woman—male/female; mature in years). These component processes—encoding, inferring, mapping, and application—are theorized to underlie any type of analogical reasoning problem, commonly included on intelligence tests.⁸ Similar analyses are being undertaken by researchers in an attempt to understand better the process demands of other information-processing tasks.⁹

The benefit of such process analysis is that educators can consider more than just the outcome or score of the instruments administered in gifted identification. Of course, there would be

no need or desire to disregard the product or score from a psychometric measure. Instead, educators would be permitted the luxury of looking beyond the score to the rich data base that existed therein. In analyzing the process demands in relation to learner performance, judgments such as the following could be made: (1) Are the process demands on learners appropriate and justified? (2) Can students' perceived difficulty in test performance be traced to underlying process deficiencies or to some more superficial condition? The failure of the individual to perform exceptionally well may reflect a simple lack of word knowledge, or it may represent a deficiency in inferring, mapping, or application. A focus on product, alone, would not consider such possibilities.

Information Analysis

While process analysis examines the processing requirements of identification instruments, information analysis investigates the knowledge base assumed within test/task items. All items on any identification measure assume some kind of background knowledge on the part of the test-taker, though the form or the depth of that knowledge will differ greatly. What the research has consistently demonstrated is that the more experience or knowledge one has about a particular subject or topic, the better one's performance.¹⁰ For example, answering questions related to a passage about baseball would be easier for someone who has played or watched baseball than answering questions on a passage about cricket, a more obscure sport in the United States.

In making identification decisions, educators must begin to consider the relationship between experience and performance. Does exceptional performance result from exceptional abilities or from exceptionally rich experiences that relate directly to tested materials? Conversely, does a less than exceptional performance arise from the testing of material that is experientially biased? The complaints of minority groups with regard to many standardized tests would serve as substantiation of this type of experiential bias from a cultural perspective. Fillmore and his colleagues have been involved in detailed analysis of the background-knowledge requirements of selected items from standardized reading tests.¹¹ Their work has supported the contention that, to perform well on many common test items, one needs more than processing abilities. A compatible experiential base is also critically important.

Response Analysis

Fillmore has also found that, even if students have exceptional processing abilities and possess the appropriate background knowledge, their response might not match the ex-

pected, "best" response. What, on the surface, appears to be an inappropriate answer may, in actuality, be a very logical, highly-inferential response. If educators never look beyond the score, the reasoning behind the responses remain untapped. Snow reiterated this position when he remarked that much relevant information can be lost to investigators if they do not take into account what is occurring inside subjects.¹² This would require the application of such techniques as interviewing, questioning, probing, and observing at various points in the testing procedure.

However, is it worth risking the predictive validity of some instruments by tampering with standardized procedures? Is the additional investment in time and effort warranted? There is no clear answer to these complex questions. Each decision must be weighed against its costs and benefits and against its theoretical justification. Yet the ambiguity of various test items and their heavy dependence on cultural and linguistic experiences remains notorious.¹³ If we must make judgments about an individual's performance outputs, then we must gather data relative to the input experiences and cognitive processes that led to those outputs.¹⁴ A test score is only one source of data, and it does not provide all the information we require to make wise decisions, especially when those decisions may affect individuals' academic futures.

To those cognizant of writings in the gifted area, this last proclamation may not seem earth-shattering. There are those who have been campaigning for more egalitarian identification procedures that rely on varied, diverse information sources.¹⁵ However, it could be argued that the use of a comprehension strategy or multiple-criteria approach in gifted identification is frequently construed to be the administration of a combination of subjective and objective instruments. What has been suggested under the information processing framework is not necessarily the administration of more instruments, but more in-depth analysis of those instruments already being employed. In addition, the call for a richer data base in identification, rather than a single score or product, differs from what may appear in the gifted literature for the very fact that it is justified by the research within an identified theoretical paradigm. Again, to say we should do something is one thing, but to demonstrate why, theoretically and empirically, is yet another.

Individualized Instruction

Within the domain of cognitive psychology, there are research programs—schema theory,¹⁶ prior knowledge,¹⁷ and study strategies¹⁸—that have shed light on individual differences in learner performance. This knowledge can be applied in developing instructional programs for gifted learners and adapting to their unique learning styles.

For example, schema theory offers insights into the way in-

formation is organized within the mind. Research in schema theory has demonstrated that the storage and retrieval from memory of critical information is facilitated by systematically structuring information during learning to demonstrate visually the relationship among ideas, a procedure called cognitive mapping. Gifted students can receive direct instruction in cognitive mapping techniques that would permit the maximized learning of content.¹⁹

Similarly, prior knowledge research explores the experiential, social, and cultural contexts involved in the learning process and evaluates their potential affect on the performance of the individual with similar or dissimilar contextual referents. Gifted students can learn to relate appropriate life experiences to the comprehension of less familiar material; in essence, relating the known to the new. Further, teachers of the gifted can design instruction to complement the socio-cultural experiences of their students, moving in small, determined steps to more diverse experiences.

The research on study strategies highlights the strategies and effort learners apply in the completion of some specified instructional task. As Bransford noted:

. . . truly active and efficient learners know what they themselves need to do in order to comprehend and master information. They know when to ask questions, when to reread, how to spend their study time, and so forth.²⁰

Even the gifted can benefit from instruction that highlights the strategies and techniques, such as lookbacks,²¹ summarizing,²² and self-questioning,²³ that culminate in improved learning.

This ability of individuals to monitor their performance during studying can be more generally identified by the label, metacognition. Metacognition is a topic that has drawn increasing attention to the investigation of cognitive processing.²⁴ It can be defined as the ability of individuals to monitor their level of understanding: knowing, knowing what one knows or does not know, and knowing what to do to remedy instances of knowledge gaps.²⁵ Among the metacognitive skills required in the successful monitoring of understanding are the development of self-regulative learning skills, the transfer of self-regulatory skills to new or varied situations, and the development of internal criteria to guide self-evaluation processes.²⁶

The purpose of highlighting these various metacognitive skills is to illustrate the point that these monitoring abilities are developed to differing degrees within different individuals. An individualized instructional program should focus on strengthening the self-regulatory processes within gifted learners. Perhaps giftedness, itself, relates in some way to those who monitor their levels of understanding exceptionally well, seek clarification

when needed, and transfer learning to new and varied situations. Certainly, the more efficient and effective learner would be more apt to stand out within the educational crowd as "gifted." There are even those who would state that intelligence could be described as one's ability to learn in the absence of complete or direct instruction, a viewpoint that would coincide with the metacognitive position.²⁷

It is these domains of cognitive processing—schema theory, prior knowledge, study strategies, metacognition—and others that may provide a more comprehensive and adequate foundation on which gifted education can reach a better understanding of individual differences. It is also these processes that are currently the focus of information processing research. We must begin to get at the heart of individual differences if we are to comprehend better the variations in performances that result on certain tests or tasks, and if we are to plan for those variations within our gifted program. We must know more about the essence of individual differences before we labor fruitlessly on the rather superficial aspects of giftedness.

Summary

The purpose of this article has been twofold: to argue for a more definitive link to theory within the activities of gifted children, and to present cognitive psychology and, more specifically, information-processing, as a theory with the utility and adaptability to have a significant impact on gifted education. It is not the objective to offer information-processing as the sole theoretical premise applicable to gifted education. To do so, would be to misrepresent the data. Even information processing, with its broad base appeal, is not without its limitations and cannot serve as the source of all needed solutions.²⁸ Information-processing, however, does have the potential to mold many of the developments in gifted education into a more consistent, coherent structure. If, as Snow prophesized, information-processing seems destined to become the next monument in intelligence theories, then it is time those in gifted education learn how to better apply this building material in the construction of more powerful and dynamic programs.²⁹

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